

Engineering Physics (2025)

Course code 25PY101-S2

M1T1

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M1U1 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

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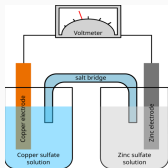
de Broglie's concept of matter waves

Heisenberg's uncertainty principle

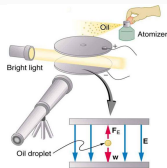
Schrödinger's time dependent wave equation

Particle in a 1D box

Macroscopic → microscopic



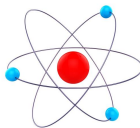
Electrochemical
cell



Oil drop
experiment



Macroscopic
Copper



Microscopic
Copper

Estimate: Avogadro number N_A

To electroplate 63.5 g of copper, it takes 2 F of charge. [Hint: 1F (F for Faraday) = 96.485 C, charge of electron $e = 1.602 \times 10^{-19}$ C]



Estimate: Radius of atom

The density of copper is 8.96 g cm^{-3} .



Faraday, Millikan

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Nature of Metal



(1)



(2)



(3)



(4), (6)



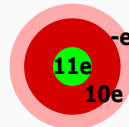
(5)

1. Lustre (Shine)
2. Solid with high 1000 °C melting points
3. Malleable (capable of being shaped)
4. Good electrical conductor
5. Good thermal conductor
6. Ductile (easy to draw wires)

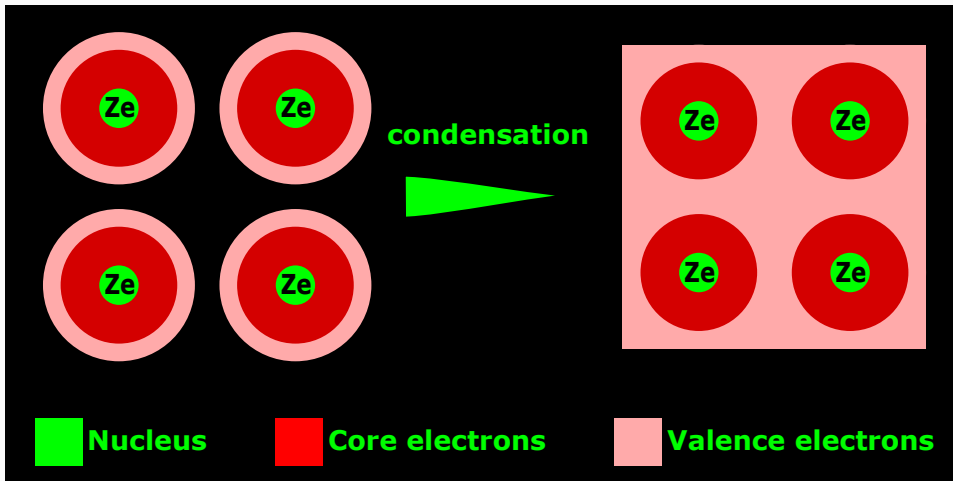
Chemistry

- Metallic bonding
- Screening

Na atom



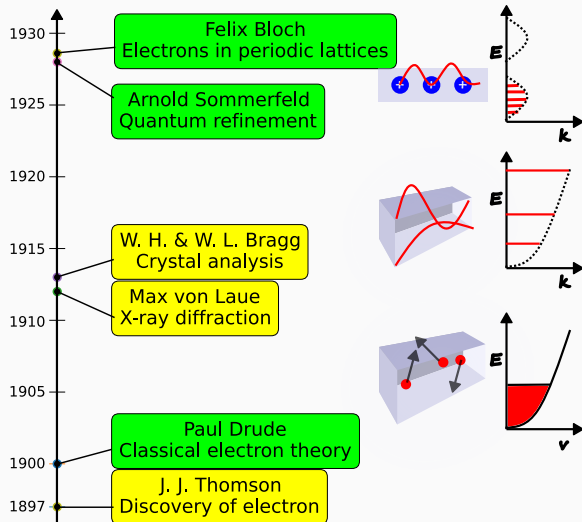
Metallic bonding $\leftrightarrow$ electron gas



Valence electrons to electron "gas"

Electron theories of metals

1. Classical free electron theory
2. Quantum free electron theory [M2 U3]
3. Quantum band theory [M2 U3]



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Classical free electron theory

- Ohm's law was discovered in 1827 and needed explanation.
- Proposed by Paul Drude in 1900 using analogy of gas of molecules.
- Mutual repulsion between electrons is neglected i.e. electrons are independent – **independent electron approximation**.
- Assumed “gas” is **free** i.e. not under influence of lattice – **free electron approximation**
- Role of lattice is to redistribute the velocity distribution that obeys **classical** Maxwell-Boltzmann statistics – classical thermodynamics.
- The theory explains Ohm's law
- Hendrik Lorentz extended theory to explain thermal conductivity – hence Drude-Lorentz electron theory.



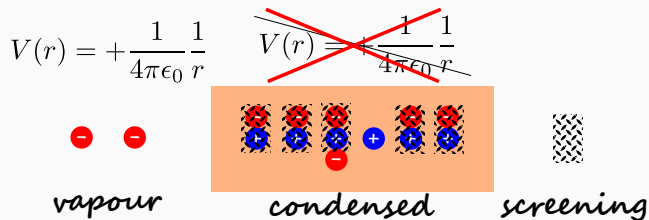
G. Ohm



P. Drude



Independent electron approximation

$$V(r) = +\frac{1}{4\pi\epsilon_0} \frac{1}{r}$$


vapour *condensed* *screening*

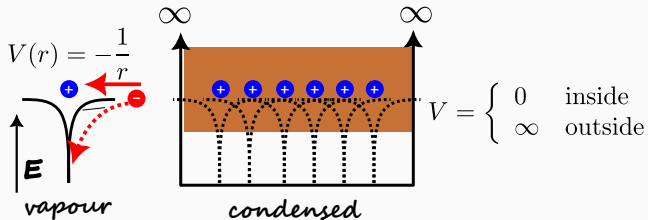
- In the condensed phase, there are of the order of 10^{23} electrons.
- The repulsive Coulomb potential between an electron and other electron is **screened** due to large numbers and results in an **independent** electron.
- The problem of motion of $\mathcal{O}(10^{23})$ electrons is reduced to the problem of motion of **single** electron.

Key Insight

The property of single electron determines the properties of electron gas.



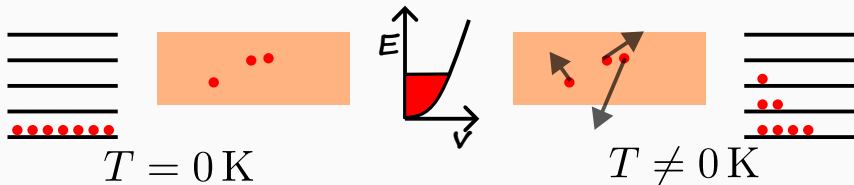
Free electron approximation



- The attractive Coulomb potential between ions and an electron is smeared due to large numbers and results in constant potential in the interior of metal with **infinite** barrier at the surface. This potential is called **infinite potential well**.
- This potential traps the electrons analogous to gas atoms in a container.
- The electrons are free to move in the potential well. The kinetic energy K of an electron with mass m and velocity $\mathbf{v}$ is

$$K = \frac{1}{2}mv^2$$

Classical thermodynamics



- At temperature $T = 0$, electrons are frozen. At $T \neq 0$, due to the large number of electrons and collisions, the kinetic energy of electrons keeps changing.
- However, the probability to find electron with lower energy is higher than the probability to find electron with higher energy. This is the essence of Maxwell-Boltzmann statistics.
- The average kinetic energy $\langle T \rangle$ at temperature is

$$\langle K \rangle = \frac{3}{2} k_B T,$$

where k_B is the Boltzmann constant.

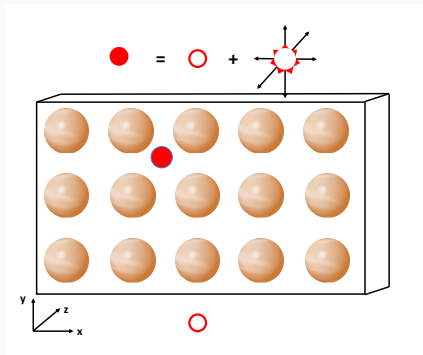


Boltzmann



Maxwell

Classical free electron theory



- Electron motion is decomposed into **thermal motion** and **drift motion**.
- Thermal motion is due to random collisions with the lattice.
- Drift motion is the net motion of electrons. The electron drifts only between collisions.

Thermal motion

Definition

Mean free path l_{mf} is defined as the average distance the electron travels between collisions.

Definition

Thermal velocity v_{th} is defined as the average velocity of the electron.

- Thermal motion has two properties mean free path l_{mf} and thermal velocity v_{th} .
- Drude assumed that l_{mf} is a material constant.
- At room temperature (300 K), the thermal velocity is of the **order of** 10^5 m s^{-1} .

Estimate: Average kinetic energy of electron

Take room temperature as 27°C . Express energy in eV.

[Hint: 1 eV = $1.602 \times 10^{-19} \text{ J}$]



Drift motion

- In the absence of external force, the drift motion is zero.
- The external force $\mathbf{F}_{ext}$ on the electron is given by the Lorentz force

$$\mathbf{F}_{ext} = -e\mathbf{E} - e\mathbf{v} \times \mathbf{B},$$

where $\mathbf{E}$ is the electric field and $\mathbf{B}$ is the magnetic field.

- The effect of the collisions is treated as a drag or frictional force

$$\mathbf{F}_{fric} = -m\frac{\mathbf{v}}{\tau}.$$

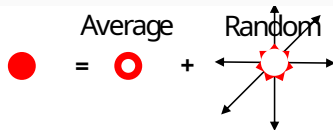
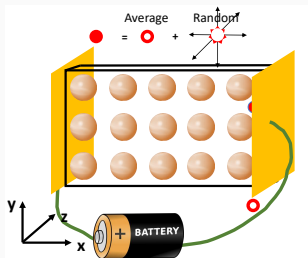
- Since Lorentz extended the theory to incorporate electromagnetic force, the theory is also the Lorentz-Drude theory.

Lorentz-Drude theory – Postulates

Postulates

1. **Particles:** Electrons are **classical** particles with mass m and charge $-e$.
2. **Kinematics:**
 - **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
3. **Dynamics:**
 - **Free electron approximation:** Electrons are free and move in an **infinite potential well**.
 - **Thermal motion:** Electron undergoes **random** collision with ion cores. Random collisions lead to thermal motion.
 - **Drift motion:** In between collisions, the Lorentz force acts on electron and leads to drift motion.
4. **Classical Thermodynamics:** The average electron energy is given by Maxwell-Boltzmann statistics.

Ohm's law



- The motion of electron is decomposed into an **average** part and the **random** part.
- The average part “drifts” with a constant velocity under electric field. The random part has **zero** net motion.

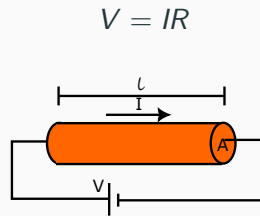
Learning Objectives

Classical electron theory explains Ohm's law qualitatively.



Ohm's law: Macroscopic version

- The **macroscopic** version of the Ohm's law states that potential V applied across a metal is directly proportional to the current I flowing across it. The constant of proportionality is called the resistance R .
- Its **microscopic** version relates the electric field and current density
- Electric field $\mathbf{E}$ is defined as the potential per unit distance. Its SI unit is V m^{-1} .
- Current density $\mathbf{j}$ is defined as the current flowing through a wire per unit area, where A is the cross-sectional area of the wire. Its SI unit is A m^{-2} .



$$V = IR$$

$$\mathbf{E} = \frac{V}{\ell} \hat{\mathbf{E}}$$

$$\mathbf{j} = \frac{I}{A} \hat{\mathbf{j}}$$

Ohm's law : Microscopic version

Lets start with the macroscopic version $V = RI$ and derive the microscopic version of Ohm's law

Derivation

Upon substitution of V, I in terms of $\mathbf{E}, \mathbf{j}$, we have

$$\frac{\ell}{V}\mathbf{E} = \frac{A}{I}\mathbf{j} \Rightarrow \mathbf{j} = \frac{\ell}{A} \frac{I}{V}\mathbf{E} \Rightarrow \mathbf{j} = \frac{\ell}{AR}\mathbf{E} = \sigma\mathbf{E}$$

Thus, the microscopic version of Ohm's law is given by

$$\boxed{\mathbf{j} = \sigma\mathbf{E}}$$

where σ is given by

$$\sigma = \frac{\ell}{AR}$$

Conductivity – σ

Definition

The conductivity σ is defined as the current density carried by the wire per unit of electric field.

$$\sigma = \frac{j}{E} \quad \text{SI unit} \quad [\sigma] = \left[\frac{\ell}{AR} \right] = \Omega^{-1} \text{m}^{-1}.$$

Question

A copper wire of length $L = 2 \text{ m}$ and diameter $d = 1 \text{ mm}$ is connected across a DC source. When a voltage of $V = 0.5 \text{ V}$ is applied across the ends of the wire, an electric current of $I = 22 \text{ A}$ is observed. Using these measurements, find the electrical conductivity σ of copper.

Key Insight

Metals have high conductivity.



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Free electron density – n

Definition

- Electron density n is defined as the number of electrons per unit volume. Its SI unit is m^{-3} .
- For metals, each atom contributes as many electrons as its **valency**. For example, Na has valency 1.

Estimate: Electron density of metals



1. Take the radius r_s of atom is nearly $3a_0$, where $a_0 = 0.529 \text{ \AA}$ is the Bohr radius.
2. Deduce n is of the order $10^{22} \text{ cm}^{-3} = 10^{28} \text{ m}^{-3}$.
3. For Cu, the experimental observed $n = 8.47 \times 10^{28} \text{ m}^{-3}$.

Key Insight

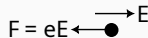
$\frac{r_s}{a_0}$ determines the free electron density.



Drift velocity – v_d

Definition

- **Drift velocity v_d** is defined as the average velocity of an electron in the presence of electric field.
- **relaxation time τ** is defined as the average time between collisions.



A diagram showing a black dot representing an electron. To its right is a horizontal arrow pointing right, labeled 'E' for electric field. To the left of the dot is a horizontal arrow pointing left, labeled 'F = eE' for the force on the electron.

Derivation

- *The equation of motion of electron is*
- *During time duration τ , electron reaches drift velocity*

$$\mathbf{a} = -\frac{e}{m}\mathbf{E}$$

$$\mathbf{v}_d = -\frac{e\tau}{m}\mathbf{E}$$

Key Insight

Drift velocity is proportional to the applied electric field.



Definition

Mobility μ is defined as the drift velocity per unit electric field

$$\mu = \frac{v_d}{E}.$$

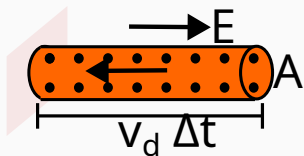
- It is a material property as

$$\mu = \frac{e\tau}{m} \quad \left[\because \mathbf{v}_d = -\frac{e\tau}{m} \mathbf{E} \right]$$

- Its SI unit is

$$[\mu] = \frac{[v_d]}{[E]} = \frac{[v_d \ell]}{[V]} = \text{m}^2 \text{V}^{-1} \text{s}^{-1}$$

Conductivity σ



In a time duration Δt , the number of electrons N in a volume of $V = v_d \Delta t A$ will pass through the cross section of area A .

Derivation

1. Current I :

$$I = -\frac{Ne}{\Delta t} = -\frac{n \cdot V}{\Delta t} = -nev_d A \quad (1)$$

2. Current density j :

$$j = \frac{I}{A} = -nev_d = \frac{ne^2 \tau}{m} E \quad (2)$$

3. Conductivity σ :

$$\sigma = \frac{ne^2 \tau}{m} = ne \left(\frac{e \tau}{m} \right) = ne \mu \quad (3)$$

Learning Objectives

We have derived the expression for conductivity.



Relaxation Time – τ

Assumption

Relaxation time τ is assumed to be a **constant**.

Estimate: Relaxation time in metals

Estimate the electron relaxation time in metals.



1. Take conductivity $\sigma \sim 10^7 \text{ S m}^{-1}$, number density $n \sim 10^{28} \text{ m}^{-3}$
2. Take electron mass $m = 9.1 \times 10^{-31} \text{ kg}$, charge $e = 1.609 \times 10^{-19} \text{ C}$.
3. Deduce relaxation time $\tau \sim \boxed{10^{-14} \text{ s or } 10 \text{ fs}}$.

- Larger $\tau \rightarrow$ fewer collisions $\rightarrow$ higher conductivity.

Trends



1. As temperature increases, τ decreases.
2. As impurity concentration increases, τ decreases.

Why?

Why?

Resistivity – ρ

Definition

Resistivity ρ is defined as the inverse of conductivity

$$\rho = \frac{1}{\sigma}$$

Its SI unit is $\Omega \text{ m}$.

- Resistivity $\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$.
- Its SI unit is $\Omega \text{ m}$.

Trends

1. As number density increases, ρ decreases.
2. As temperature increases, ρ increases.



Why?

Why?

Summary of lecture

- Derived $\sigma = \frac{ne^2\tau}{m}$.
- Connected microscopic electron properties to macroscopic Ohm's law.
- Classical free electron theory has qualitative and quantitative agreement with experiment values of conductivity.
- However, the theory has drawbacks. Some of them are
 - Incorrect prediction for conductivity vs valency.
 - Cannot explain anomalous sign of Hall coefficient in some metals.
 - Underestimation of mean free path.
 - Cannot explain classification of materials into conductors, semi-conductors and insulators.
 - Wrong prediction of conductivity vs temperature.
 - Overestimation of heat capacity.

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Semiconductors: Classification

Definition

Semiconductor is a class of condensed matter having electrical conductivity (10^{-4} S m^{-1} to 1 S m^{-1}) greater than that of insulator ($<10^{-10} \text{ S m}^{-1}$) but lower than that of conductor (10^7 S m^{-1}).

- In the periodic table, there are 11 elements that are semiconductors. These are called **elemental semiconductors**. Si and Ge from Group IV are widely used elemental semiconductors.
- Certain combinations of elements of Groups III and V or Groups II and VI also are semiconductors. These are called **compound semiconductors**. GaAs, InP are some of the well known examples.
- In a semiconductor, two charge carriers, namely **electrons** and **holes**, transport current.

Intrinsic Semiconductor

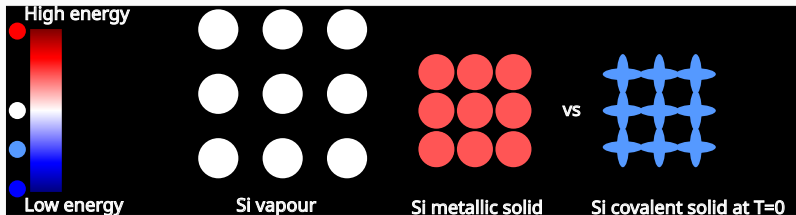
- Chemically pure semiconductor is called intrinsic semiconductor.
- Practically, a semiconductor is considered pure if there is less than one impurity atom per billion host atoms i.e. < 1 ppb (part per billion).
- It has five properties
 1. Perfect insulator at absolute zero temperature
 2. Generation of charge carriers by thermalization
 3. Existence of energy band gap (M2U3)
 4. Conductivity is **highly** influenced by temperature
 5. Recombination of charge carriers
- For Si, the valence electron density is of the order of 10^{29} m^{-3} . However, at room temperature the conduction electron density is only of the order of 10^{18} m^{-3} .

Estimate: Valence electron density of Si

The lattice constant of Si crystal is $5.43 \text{ \AA}$.



Nature of intrinsic semiconductor at $T = 0$ (property 1)



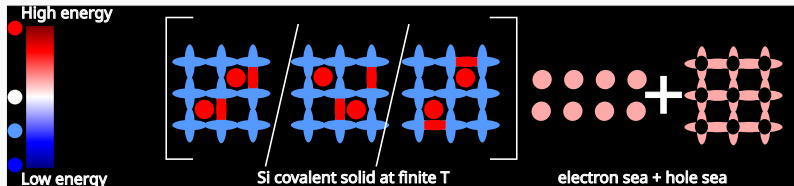
- Valence electrons are strongly attracted to the nucleus. This prevents the formation of free electron gas.
- Instead the valence electrons are shared between neighbouring atoms resulting in **covalent bond**.
- As all the valence electrons are involved in bonding,
there are no “free” electrons at absolute zero temperature.

Key Insight

Intrinsic semiconductor is a perfect insulator at absolute zero!



Nature of intrinsic semiconductor at $T \neq 0$ (property 2)



- However, at finite temperature, due to thermalization, few bonds are broken, releasing the electrons to the **free electron sea**.
- In the process of bond breaking, the crystal lattice deficient with one electron can be considered as a new type of charge carrier. This is defined as the **hole**.
- The holes are released into the **free hole sea**.
- Every free electron is associated with a free hole. Hence they are called **electron-hole pair**.

Definition

An intrinsic semiconductor is a semiconductor crystal in which the electrical conduction arises due to **thermally excited** electrons- hole pairs.

Nature of hole: Drift motion (property 2)



(a) Analogy for a hole: Bubbles (b) Drift motion of holes

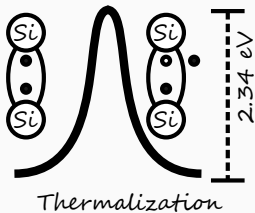
- A hole is a **quantum vacancy** that is described as an absence of electron.
- Analogy is the bubbles appearing in carbonated water. The bubbles are absence of water.
- Under application of electric field, the bonded electrons move in the opposite direction.
The hole appears to move along the electric field.

Key Insight

The motion of hole is along the electric field.



Energy band gap: Chemistry view (property 3)



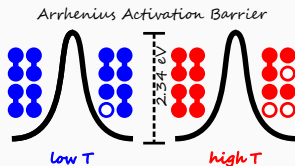
- The electron of the covalent bond needs to overcome bonding strength to become free.
- If all the bonds are broken, then the crystal vapourizes. The energy needed to vapourize a crystal is called **cohesive energy**.
- The actual energy to free electron is 1.12 eV and is called the **energy band gap** (M2U3).

Estimate: Bonding strength

The cohesive energy of Si crystal is 450 kJ mol^{-1} . Estimate the Si–Si bond strength.



Charge carriers vs temperature (property 4)



- The scale of thermal energy is given by $k_B T$. At room temperature, the thermal energy of electron is of the order of 26 meV. [Estimate]
- The bond strength can be considered an **activation barrier** for the generation of electron-hole pair. This is analogous to rate of chemical reaction

covalent bond + thermal energy $\rightarrow$ electron + hole

- Electron-hole pair density increases exponentially with temperature. This follows the **Arrhenius law**.

$$n \propto \exp\left(-\frac{1}{k_B T}\right) \quad p \propto \exp\left(-\frac{1}{k_B T}\right)$$

Recombination of carriers (property 5)

- We have seen thermal energy to generate electron-hole pairs. This is an example of **equilibrium** generation
- Light can also be used to generate electron-hole pairs.

covalent bond + light $\rightarrow$ electron + hole

However, this is a **non-equilibrium** generation process.

- The reverse process where the electron-hole pairs combine to become part of the covalent lattice is called **recombination**.

electron + hole $\rightarrow$ covalent bond + light/lattice vibration

The additional energy is released either as light (**photon**) or lattice vibration (**phonon**).

- The generation-recombination of charge carriers is a **dynamic** two way process.

Key Insight

For analysis, a semiconductor is a collection of electron, holes, photons and phonons and the interaction between them.



Extrinsic semiconductor

- The electrical properties of intrinsic semiconductor can be altered by adding impurities.
- The technique of adding controlled impurity is called **doping** and the impurity added is called **dopant**.
- Doped semiconductor is called **extrinsic** semiconductor.
- Dopant atom substitutes the position of parent atom. The dopants are of two types – donor and acceptor.
- For Si, donors are P, As, etc and acceptors are B, Al, etc.
- For P as a donor dopant, four valence electrons contribute to covalent bonding. The fifth electron is loosely bound to the phosphorus atom
- For B as an acceptor dopant, three valence electrons contribute to covalent bonding.

Estimate: Dopant concentration

P doping is 1 ppm.



$$n_{\text{valence}} \sim 2 \times 10^{23} \text{ cm}^{-3} / 165$$

Ionization energy of dopants

Definition

The energy required to ionize a dopant atom is called ionization energy.

- Lattice vibrations (phonons) supply the ionization energy (I.E.)
- For P as donor dopant,



- Similarly, for B as acceptor dopant,



- If thermal energy (E_{th}) is greater than I.E. E_{ion} , then ions are completely ionized. This is called **complete** ionization.
- For $E_{th} \ll E_{ion}$, there is no ionization. This is called **freeze out**.
- **Partial** ionization occurs when $E_{th} \lesssim E_{ion}$
- In Si, $E_{ion} \sim 25 \text{ meV}$ and $E_{th} \sim 25 \text{ meV}$ at room temperature. Thus, the ions are **completely ionized**.

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Conductivity

In a semiconductor, there are two types of charge carriers – electrons and holes. Each carrier is associated with corresponding mobility. Thus, the mobility of electrons is μ_e and that of holes is μ_h .

From the classical free electron theory, the conductivity in terms of mobility and charge density is given by

$$\sigma = ne\mu$$

In semiconductors, there are **two channels** for conduction. Thus, the conductivity adds up and is given by

$$\sigma = ne\mu_e + pe\mu_h$$

Key Insight

Both electrons and holes contribute to conductivity.



Drift motion of electrons and holes

- Under electric field, electron and hole drift motion is **anti-parallel**.

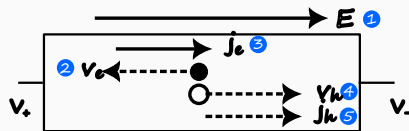
$$\mathbf{F}_e = -q_e \mathbf{E}, \quad \mathbf{F}_h = +q_h \mathbf{E}.$$

- The sign of their charges are opposite.
- Electron and hole drift current direction is **parallel**.
- The total drift current density is given by
- Drift current is proportional to electric field.
- Conductivity is written in terms of carrier density and mobility

$$\sigma_e = ne\mu_e, \quad \sigma_h = pe\mu_h,$$

- The drift conductivity is given by

$$\sigma_{\text{drift}} = ne\mu_e + pe\mu_h$$



$$\mathbf{j}_{\text{drift}} = \mathbf{j}_e + \mathbf{j}_h.$$

$$\mathbf{j}_{\text{drift}} = \sigma_e \mathbf{E} + \sigma_h \mathbf{E}$$

Intrinsic semiconductor

For the intrinsic case, the number density of electrons equals the number density of holes

$$n = p = n_i$$

Thus, the conductivity is given by

$$\sigma = n_i e (\mu_e + \mu_h)$$

Estimate: Conductivity of intrinsic Si

Compare it with conductivity of copper.

Key Insight

Though the carrier mobility in metals is less than in semiconductors, **very much higher** carrier density leads to higher metallic conductivity.

Carrier density	cm ⁻³
n_{i-Si}	10^{10}
n_{Cu}	10^{22}
Mobility	cm ² V ⁻¹ s ⁻¹
$\mu_e [Si]$	1400
$\mu_h [Si]$	500
$\mu [Cu]$	40



Extrinsic semiconductor

- n-type: Electrons are **majority** charge carriers

$$n \gg p, \quad \text{so that} \quad ne\mu_e \gg pe\mu_h$$

$$\therefore \sigma \simeq ne\mu_e$$

Estimate: Conductivity of extrinsic n-type

Si



P doping is $5 \times 10^{16} \text{ cm}^{-3}$. Compare it with conductivity of intrinsic-Si.

Key Insight



Doping of the order of 1 ppm increases conductivity of silicon by six orders of magnitude.

- p-type: Holes are **majority** charge carriers

$$p \gg n, \quad \text{so that} \quad pe\mu_h \gg ne\mu_e$$

$$\therefore \sigma \simeq pe\mu_h$$

Material	n (in cm^{-3})	σ (in S m^{-1})
i-Si	10^{10}	10^{-4}
1 ppm n-Si	10^{16}	10^2

M1U1 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

Hall effect

- In the presence of magnetic field acting perpendicular to the flow of electrons, the electrons experience a Lorentz force that deviates their path in the transverse direction. Accumulation of electrons along the edges leads to generation of voltage. This is called **Hall effect**.



E. Hall

- Hall effect is prominent in two dimensional geometry and can be considered the advent of research on thin films.
- It was discovered before the discovery of electron!
- In the context of semiconductors, Hall effect has contribution from holes, in addition to electrons. It also helps determine
 - the type of semiconductor (n or p),
 - the majority carrier concentration, and
 - the majority carrier mobility.

Hall effect: Lorentz force

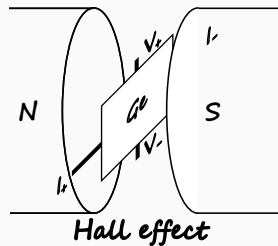
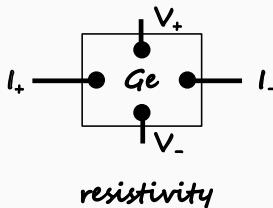
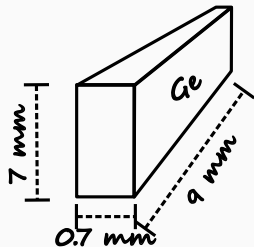
- In the presence of electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$, the electron experiences a force given by

$$\begin{aligned}\mathbf{F}_{\text{Lorentz}} &= \mathbf{F}_{\text{elec}} + \mathbf{F}_{\text{mag}} \\ &= -e\mathbf{E} - e\mathbf{v} \times \mathbf{B}\end{aligned}$$



H. Lorentz

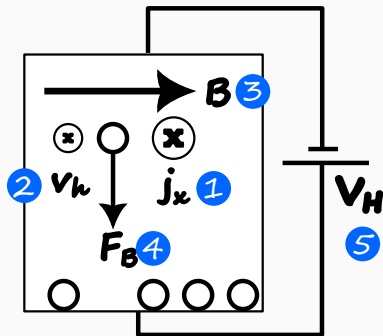
Hall effect: experiment



- The aim is to find the charge carrier type of Germanium crystal of dimensions 9 mm x 7 mm x 0.7 mm.
- The resistivity is measured in the **four-probe geometry**.
- Current is applied along x direction.
- Magnetic field of 0.1 T is applied along z direction.
- The Hall voltage is measured along y direction.

$$[1 \text{ T} = 10^4 \text{ G}]$$

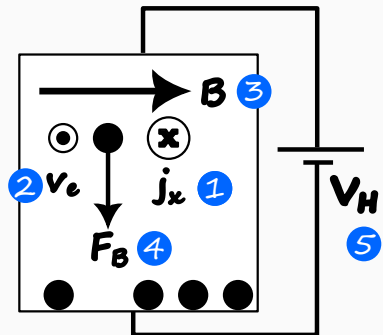
Hall effect in p-type semiconductor



Hall effect for holes

1. Current along x .
2. Electron drift along x .
3. Magnetic field along z .
4. Magnetic force along $-y$.
5. The buildup of holes **induces** an electric field in the $+y$ direction.
6. The corresponding voltage across the width is positive.

Hall effect in n-type semiconductor



Hall effect for electrons

1. Current along x .
2. Electron drift along $-x$.
3. Magnetic field along z .
4. Magnetic force along $-y$.
5. The buildup of electrons **induces** an electric field in the $-y$ direction.
6. The corresponding voltage across the width is negative.

Hall field and Hall voltage

- In the steady state, the magnetic field force is balanced by the induced electric field

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = q(E_y - v_x B_z) = 0$$

- The induced electric field, called the **Hall field** is given by

$$E_y = v_x B_z.$$

- The Hall field produces a voltage across the semiconductor, which is called the **Hall voltage** given by

$$V_H = E_H w$$

where w is the width of the semiconductor.

- The Hall voltage is positive for p-type and negative for n-type semiconductor.

Hall coefficient

Definition

Hall coefficient R_H is defined as the Hall field per unit current density per unit magnetic field.

- R_H is given by

$$R_H := \frac{E_H}{j_x B_z} = \frac{v}{j_x}$$

- E_H is positive for holes and negative for electrons.
- However, the current density j is related to the drift velocity v by

$$j = nev \quad [j = pev \quad \text{for holes}]$$

- Therefore,

$R_H(\text{electron}) = -\frac{1}{ne}, \quad R_H(\text{hole}) = +\frac{1}{pe}$
--

Charge carrier properties

- Semiconductor type: The sign of Hall voltage determines the type of majority carrier in a semiconductor
- Majority carrier density: The Hall coefficient gives the majority carrier density as

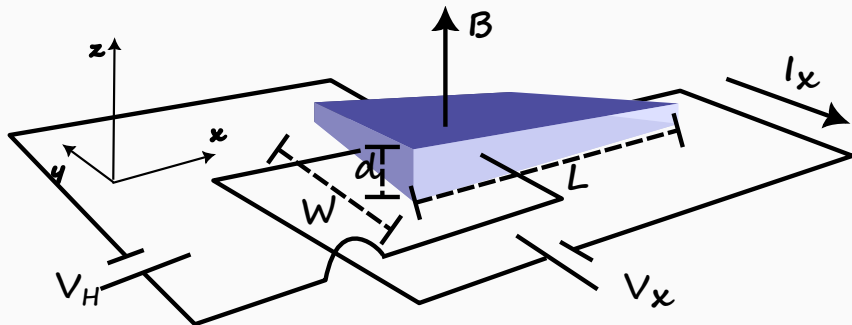
$$n = \frac{1}{R_H e}$$

- Majority carrier mobility: If the conductivity of the semiconductor σ is measured in the absence of magnetic field, then the mobility of majority carriers is given by

$$\mu = \frac{\sigma}{ne}$$

Test your Understanding

Problem



Determine the majority carrier type, concentration and mobility. $L = 10^{-1}$ cm, $W = 10^{-2}$ cm, $d = 10^{-3}$ cm. $I_x = 1$ mA, $V_x = 12.5$ V, $B = 500$ G, $V_H = -6.25$ mV.

M1U1 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

M1U2 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

M1U2 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

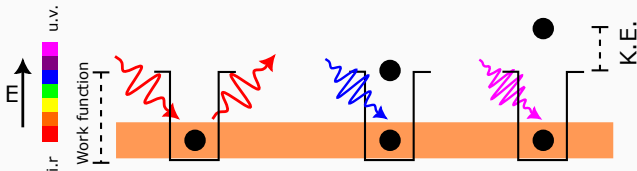
Schrödinger's time dependent wave equation

Particle in a 1D box

Photo-electric effect

Definition

Light of **suitable** frequency leads to **instantaneous** emission of electron from a metal surface.

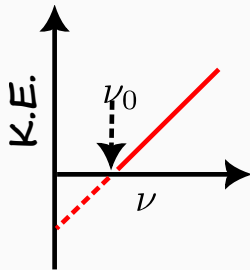


A. Einstein

- Discovered by Albert Einstein in 1905.
- The emitted electron is called the **photo-electron**.
- The minimum energy required to emit an electron is a material property and is called the **work function** of the metal.

Classical explanation of photo-electric effect

- Light is a wave with energy given by its intensity.
- Therefore, even low-frequency light (if sufficiently intense) should eventually provide enough energy to eject electrons — though perhaps after some time delay.



- But it is observed that
 - Emission of electrons is instantaneous – even for low intensity light as long as frequency of light is above a certain threshold ν_0 .
 - No electrons are emitted if the frequency is below the threshold, no matter how intense the light is.
 - The kinetic energy of emitted electrons depends on the frequency of light, not its intensity.

Key Insight

Classical electromagnetic theory cannot explain photoelectric effect!

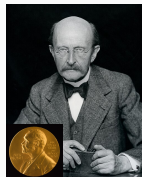


Quantum mechanical explanation of Photoelectric effect

- Albert Einstein borrowed the idea of **quantum**¹ from Max Planck's theory of blackbody radiation and applied to photoelectric effect.

Planck's law

- Max Planck postulated that the light energy is absorbed or emitted by matter in is a packet of energy. The quantum of energy is related to the frequency ν by $E = h\nu$ where h is the **Planck's constant** given by 6.625×10^{-34} J s.
- The quantum of light energy is called **photon**.



Max Planck

Estimate: Energy of infra-red photon

Wavelength of infra-red photon is $1 \mu\text{m}$.

¹“Quantum” is Latin for “a certain amount”. In the context of physics, the word quantum refers to a packet.



[Hint: $c = \nu\lambda$]

Photoelectric effect: energy equation

- Photon with energy $E = h\nu$ transfers energy to the electron, to overcome the work function Φ . The rest of the energy is photoelectron's kinetic energy.

$$\text{K.E.} = h\nu - \Phi$$

- The photoelectrons generate photocurrent.
- In the experiment, the photo electron emitted is decelerated by applying a **stopping potential** V_{sp} till the photocurrent is zero. Thus,

$$\text{K.E.} = eV_{sp}$$

- Therefore, frequency is related to stopping potential as

$$eV_{sp} = h\nu - \Phi$$

- The plot of $\nu(\rightarrow x)$ vs $V_{sp}(\rightarrow y)$ is a straight line $y = mx + c$ with slope $m = h/e$ and y-intercept $c = -\Phi$.
- The x-intercept is the minimum frequency of photon to generate photo-current and is called the **threshold frequency** $\nu_0 = \frac{\Phi}{h}$.

Photoelectric effect: problems

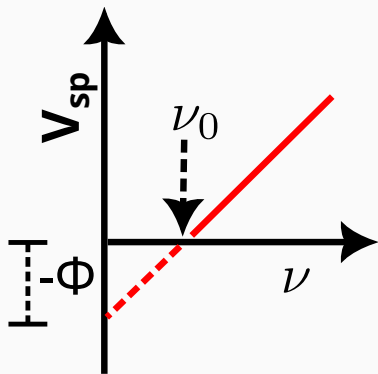
Estimate: Energy of X-ray photon

Wavelength of X-ray photon is $0.708 \text{ \AA}$.



[Hint: (a). $E \propto 1/\lambda$. (b). $E_{\lambda=1 \mu\text{m}} = 1.24 \text{ eV}$]

Problem



The work function Φ of Au is 4.90 eV .

1. A blue photon is incident on Au. Does it emit photo-electron? If yes, what is the stopping potential required to have zero photocurrent.
[$\lambda_{\text{blue}} = 460 \text{ nm}$]
2. Determine the threshold frequency ν_0 for photon emission of electrons for Au.

Repeat the above steps for Cs with work function 1.90 eV .

M1U2 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

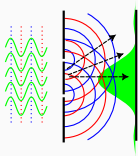
de Broglie's concept of matter waves

Heisenberg's uncertainty principle

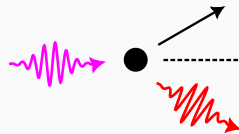
Schrödinger's time dependent wave equation

Particle in a 1D box

Radiation: Wave vs Particle

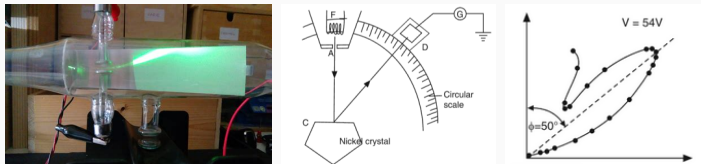


- Young's double slit experiment
- Discovered in 1801
- Light is a wave with amplitude, frequency and wavelength
- **Principle of superposition:** Waves superpose leading to constructive and destructive fringes.
- Animation for double slit experiment [\[Click\]](#)



- Compton effect
- Discovered in 1923.
- Light is a particle with momentum!
- **Conservation of momentum:**
Light redshifts (wavelength increases) upon scattering off an electron.

Matter: Particle vs Wave



(a) cathode ray tube, Davison-Germer experiment (b) and data (c).

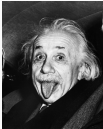
- Thomson's cathode ray tube experiment
- Discovered in 1897
- Electron is a particle
- Davisson and Germer experiment
- Discovered in 1927.
- Electron is a wave!

Key Insight

The name of the game is light-matter interaction. And nature plays the game in a **symmetric** fashion!



Summary of nature of light-matter interaction

	Wave	Particle
Light	 Young's double slit ²	  Einstein's photo-electric effect Compton's scattering effect
Electron	 Davison's and Germer's diffraction experiment	 Thomson's cathode ray tube experiment

²British polymath who has been described as “The Last Man Who Knew Everything” and disproved Newton's corpuscular theory of light.

M1U2 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

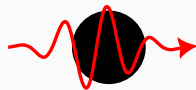
Schrödinger's time dependent wave equation

Particle in a 1D box

Wave-Particle Duality

de Broglie hypothesis

Any moving particle is associated with a wave. The waves associated with particles are called **de Broglie waves** or **matter waves**.



- Louis de Broglie in 1924 postulated the existence of **matter waves**.
- He suggested since light wave exhibits particle like behaviour, matter particles should be expected to show wave-like properties. This is the hypothesis of **wave-particle duality**.
- The wavelength λ of the matter wave is related to the momentum p of the matter particle by

$$\lambda = \frac{h}{p}$$



Matter wave: properties

- Matter waves are due to motion of particle and are independent of charge. Therefore, they are neither electromagnetic waves nor acoustic waves. They are a new kind of waves.
- Can propagate/travel through vacuum and do not require any medium for propagation.
- Smaller the mass, longer the de Broglie wavelength.
- Smaller the velocity, longer the de Broglie wavelength.
- Velocity of matter waves depends on velocity of particle and is not a constant quantity.

Estimate: de Broglie wavelength of thermal electron

Estimate at room temperature.



Estimate: de Broglie λ of macroscopic particle

Cricket ball of mass 150 g is bowled at 140 km h⁻¹.



M1U2 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

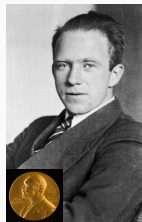
Heisenberg uncertainty principle

Uncertainty principle/hypothesis

Precise values of position and momentum of a particle cannot be determined **simultaneously**. If the uncertainty in position is Δx and uncertainty in momentum is Δp , then

$$\Delta x \Delta p \geq \frac{\hbar}{2}, \quad \text{where} \quad \hbar = \frac{h}{2\pi}$$

where $\hbar$ is the reduced/modified Planck's constant.



- Werner Heisenberg postulated the uncertainty principle in 1927.
- In classical mechanics, position and momentum are **deterministic**. In quantum mechanics, they are not deterministic and are associated with **uncertainties**.
- If position is measured with low uncertainty, then measurement of momentum has high uncertainty and vice versa.

Uncertainty → Probability

- The uncertainty principle relates any pair of **conjugate** variables.
- Position – momentum is a pair of conjugate variables.
- Energy – time is also a conjugate variable pair.
- So the uncertainty in energy ΔE is related to the uncertainty in time by Δt by

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

This is the second statement/corollary of Heisenberg's uncertainty principle.

- Due to the smallness of reduced Planck's constant, the uncertainty principle is significant for subatomic particles.
- The cause of uncertainty is **limitation of measurement**.

Key Insight

Since the position cannot be measured with certainty, we determine the **probability** of finding an electron at a particular position.



Uncertainty in function of variable

- If uncertainty in variable x is Δx , then the uncertainty in variable $f(x)$ is given by

$$\Delta f = \frac{df}{dx} \Delta x$$

- Kinetic energy E is related to momentum p by

$$E = \frac{p^2}{2m}$$

- If uncertainty in momentum is Δp , then uncertainty in kinetic energy E is given by

$$\Delta E = \frac{dE}{dp} \Delta p = \frac{p}{m} \Delta p$$

Problem

- *Uncertainty in position of electron is $12 \text{ \AA}$.*
- *Nominal energy of electron is 16 eV .*
- *Determine the uncertainty in momentum and kinetic energy*

M1U2 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

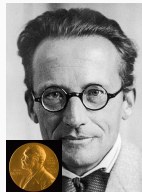
Schrödinger's wave equation

Wave equation hypothesis

If the **wave function** of a particle of mass m is $\Psi(x, t)$, then it satisfies the **wave equation** given by

$$-\frac{\hbar^2}{2m} \cdot \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x)\Psi(x, t) = j\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

where $V(x)$ is the potential function^a and j is the imaginary constant $\sqrt{-1}$.



^aPotential energy and not electric potential difference

- Erwin Schrödinger postulated the wave equation in 1926 that incorporated principles of quanta introduced by Planck and wave-particle duality introduced by de Broglie.
- The wave function $\Psi(x, t)$ describes the behaviour of particle
- $\Psi(x, t)$ can be a **complex** quantity.

Solution: Time dependent part of W.F

$$j\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = \eta$$

- The time dependent side of W.E. is then given by

$$\frac{d\phi(t)}{dt} = -j\frac{\eta}{\hbar}\phi(t) \quad \text{related to} \quad \frac{dy}{dx} = cx$$

- The solution is a sinusoidal wave given by

$$\phi(t) = e^{-j\left(\frac{\eta}{\hbar}\right)t} \quad \text{related to} \quad y = e^{-j\omega t}$$

with angular frequency ω given by

$$[\nu = \frac{\omega}{2\pi}]$$

$$\omega = \frac{\eta}{\hbar}, \quad \Rightarrow \eta = \hbar\omega = \frac{h}{2\pi} \cdot 2\pi\nu = h\nu$$

- From Planck's law, $E = h\nu$. Therefore,

$$\eta = E, \quad \text{and} \quad \phi(t) = e^{-j\left(\frac{E}{\hbar}\right)t} = e^{-j\omega t}$$

Solution: Time independent part of W.F

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + V(x) = \eta$$

- The time independent portion can now be written as

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + V(x) = E$$

where the separation constant is replaced by the total energy E .

- Therefore, the time independent part of W.E. is

$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V(x)) = 0$$

- This is called the time independent Schrödinger wave equation. The aim is to solve for the time independent wave function $\psi(x)$ for a given potential $V(x)$.

Physical meaning of wave function

Probability density hypothesis

If the **wave function** of a particle is $\Psi(x, t)$, then the probability density function is given by

$$|\Psi(x, t)|^2$$



- Max Born in 1926 postulated the **interpretation** of wave function.
- Let us assume that we have solved for the time independent part of wave function. Thus,

$$\Psi(x, t) = \psi(x)e^{-j\omega t}$$

- Since $\Psi(x, t)$ is a complex function, it cannot represent physical quantity. Its modulus squared is the **probability density** of finding particle between x and $x + dx$ and is given by

$$|\Psi(x, t)|^2 = \Psi(x, t) \cdot \Psi^*(x, t) = \psi(x)e^{-j\omega t} \cdot \psi^*(x)e^{+j\omega t} = |\psi(x)|^2$$

- The probability of finding particle between x and $x + dx$ is then

$$|\psi(x)|^2 \cdot dx$$

Boundary conditions

- Total probability is unity.

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

This condition is called **normality condition**.

- Continuity condition
 1. $\psi(x)$ must be finite, single-valued and continuous.
 2. $\frac{\partial \psi(x)}{\partial x}$ must be finite, single-valued and continuous.
- $\psi(x)$ must vanish at infinity.

$$\lim_{x \rightarrow \infty} \psi(x) = 0, \quad \lim_{x \rightarrow -\infty} \psi(x) = 0$$

This condition is due to the **physicality condition**.

Problem

1. Consider the wave function $\Psi(x, t) = A \cos\left(\frac{\pi x}{2}\right) e^{-j\omega t}$ for $-1 \leq x \leq +3$. Determine A so that $\int_{-1}^{+3} |\Psi(x, t)|^2 dx = 1$.

Summary: Postulates of quantum mechanics



The work of Planck, de Broglie, Heisenberg, Born, and Schrödinger can be summarized as

1. The state of a particle is given by the wave function $\Psi(x, t)$.
2. The probability density of the particle is $|\Psi(x, t)|^2$. The probability of finding the particle in between x and $x + dx$ is $|\Psi(x, t)|^2 dx$.
3. If the uncertainty in position is Δx and uncertainty in position is Δp , then $\Delta x \Delta p \geq \frac{\hbar}{2}$.
4. The de Broglie wavelength of particle with momentum p is $\lambda = \frac{h}{p}$.
5. The energy E of the particle with frequency ν is $E = h\nu$.



M1U2 Plan

Condensed matter

Metals

Classical free electron theory

Expression of electrical conductivity

Introduction to Semiconductors

Electrical conductivity of semiconductors

Hall effect

Concept of Panchabhuta

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

Applications of Schrödinger's W.E.

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi(x) = 0 \quad \Psi(x, t) = \psi(x)e^{-j\omega t}$$

- The aim is to solve for the time independent wave function $\psi(x)$ for a given potential $V(x)$.
- The potential functions that are useful to describe commonly occurring physical systems are

Applications of Schrödinger's W.E. (contd...)

$V(x)$	Name	Physical systems
0	Constant potential	Particle in free space
$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$	Infinite potential well	Ideal atom
$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$	Finite potential well	Work function
$V(x) = \begin{cases} V_0 & -\frac{a}{2} < x < \frac{a}{2} \\ 0 & \text{otherwise} \end{cases}$	Finite barrier	Nuclear fission Scanning Tunneling Microscope

Constant potential: problems

Problem

An electron in free space is described by a plane wave given by

$\Psi(x, t) = A \exp(j(kx - \omega t))$. If $k = 8 \times 10^8 \text{ m}^{-1}$ and $\omega = 8 \times 10^{12} \text{ rad s}^{-1}$, determine

- 1. phase velocity and wavelength of the plane wave*
- 2. momentum and kinetic energy (in eV)*

Repeat the steps for $k = -1.5 \times 10^9 \text{ m}^{-1}$ and $\omega = 1.5 \times 10^{13} \text{ rad s}^{-1}$.

Problem

Determine the wave number, wavelength, angular frequency, and period of wavefunction that describes an electron traveling in free space at a velocity of (a) $5 \times 10^6 \text{ cm s}^{-1}$.

Summary of particle in free space

- For a particle in free space, all wave vectors are allowed $k = -\infty$ to $+\infty$.
- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

- The time independent part of wave function is given by

$$\psi(x) = e^{jkx}$$

- The time dependent part of wave function is given by

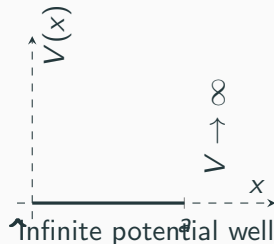
$$\phi(t) = e^{-j\omega t}, \quad \text{where} \quad \omega = \frac{E}{\hbar}$$

- The total wave function is given by

$$\Psi_k(x, t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx - \omega t}}.$$

$V(x)$: Infinite potential

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$$



- Consider a particle in a 1-dimensional infinite potential well.
- The well can be considered as a box.

Outside infinite box

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - \infty)\psi(x) = 0$$

Inside infinite box

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

$V(x)$: Infinite potential well – Solution of TISE

Outside infinite box

- There is no wave function!

$$\psi(x) = 0$$

- Upon applying boundary conditions

Left B.C.

$$\psi(x = 0) = 0$$

$$A \cos kx = 0$$

$$\Rightarrow A = 0$$

Inside infinite box

- The wavefunction is

$$\psi(x) = A \cos kx + B \sin kx$$

Right B.C.

$$\psi(x = a) = 0$$

$$B \sin ka = 0$$

$$\Rightarrow \sin ka = 0$$

$V(x)$: Infinite potential well – Solution of TISE

- The equation $\sin ka = 0$ is valid if

$$ka = n\pi$$

where n is a positive integer, or $n = 1, 2, 3, \dots$. The parameter is referred to as quantum number. In terms of the quantum number, the wave vector is given by

$$k = \frac{n\pi}{a}$$

- The prefactor B can be found by applying the **normality condition** of wave function.

$$\int_0^a |B \sin kx|^2 dx = 1, \quad \Rightarrow \quad \int_0^a B^2 \sin^2 kx dx = 1$$

$$\therefore B = \sqrt{\frac{2}{a}}$$

- Therefore, the time independent wave function is given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$V(x)$: Infinite potential well – Energy levels

- Recall that the wave vector k is related to the energy E by

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \Rightarrow \quad E = \frac{\hbar^2 k^2}{2m}$$

- Therefore, the energy is given by

$$E = E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

Problem

Electron in an infinite potential well of width 5 Å. Calculate the first three energy levels.

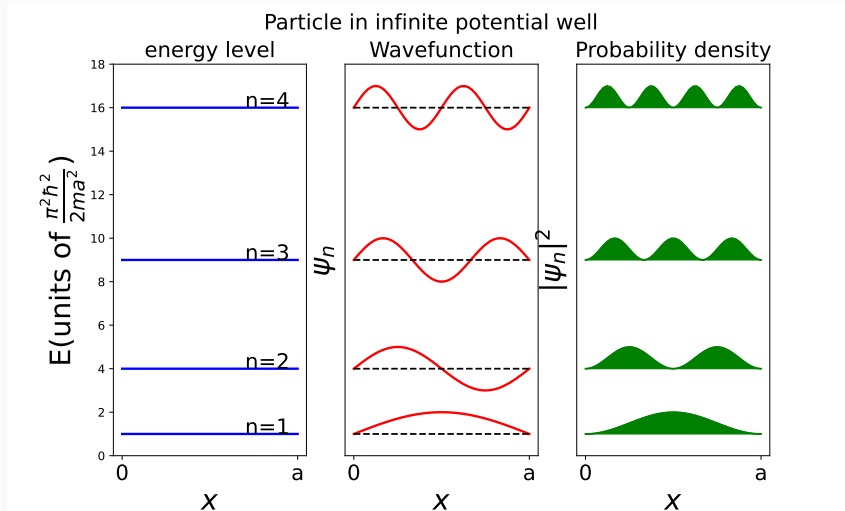
Key Insight

The energy of particle is **quantized** i.e. it can only have particular **discrete** values.



$V(x)$: Infinite potential well – Nature of solutions

- The solutions are **standing** waves.



1D infinite well: problems

Problem

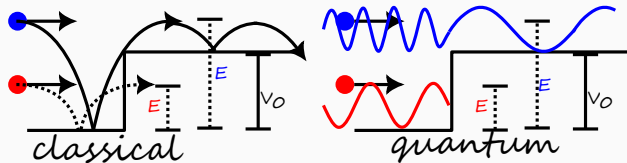
1. *A particle with mass of 15 mg is bound in a 1D infinite potential well that is 1.2 cm wide.*
2. *If the energy of the particle is 15 mJ, determine the value of n for that state.*
3. *What is the energy of the $(n + 1)$ state?*
4. *Would quantum effects be observable for this particle?*

Summary of particle in infinite potential well

For a particle in an infinite potential well,

Description	Equation
Discrete wave vectors	$k_n = \frac{n\pi}{a}$.
Energy levels	$E_n = \frac{\hbar^2 k^2}{2m}$, or $E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$
Time independent part of wavefunction	$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$
Time dependent part of wave function	$\phi_n(t) = e^{-j\omega_n t}$, where $\omega_n = \frac{E_n}{\hbar}$
Total wave function	$\Psi_n(x, t) = \psi_n(x)\phi_n(t)$,

Potential energy barrier – Classical vs Quantum



- There are two regions – (I) without barrier and (II) with barrier
- Region I**

- Total energy E is kinetic energy T_1
- Wave vector is real
- Wave function has sinusoidal form

$$T_1 = E$$

$$k_1 = \sqrt{\frac{2mT_1}{\hbar^2}}$$

$$\psi_1(x) = A \exp jk_1x + B \exp -jk_1x$$

Region II

- There are two cases – (i) total energy is greater than potential energy barrier and (ii) total energy is lesser than potential energy barrier.

Case (i) – $E > V_0$

- Classically, the particle overcomes the potential barrier with decreased kinetic energy.
- Quantum mechanically, the particle overcomes the potential barrier with decreased wave vector.

Region II

- Kinetic energy is reduced by V_0
- The wave function is given by

$$T_{II} = E - V_0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m(E - V_0)}{\hbar^2}\psi(x) = 0$$

- Wave vector is real

$$k_{II} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}$$

- Wave vector is reduced compared to region I
- Wave function has sinusoidal form

$$\psi_{II}(x) = A \exp jk_{II}x + B \exp -jk_{II}x$$

Curious case – $E < V_0$

- Classically, the particle **cannot** overcome the potential barrier.
- However, quantum mechanically, it is possible!
- Negative kinetic energy is allowed in quantum mechanics.

Region II

- Total energy E is less than potential barrier
- Kinetic energy is negative!
- Wavevector is imaginary

$$T_{II} = E - V_0 < 0$$

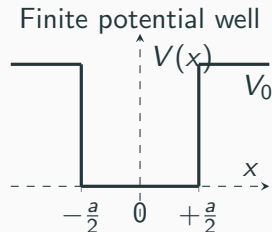
$$k_{II} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} = j\sqrt{\frac{2m(V_0 - E)}{\hbar^2}} = j\kappa_{II}$$

- Wave function has exponential form

$$\begin{aligned}\psi_{II}(x) &= A \exp jk_{II}x + B \exp -jk_{II}x \\ &= A \exp -\kappa_{II}x + B \exp +\kappa_{II}x\end{aligned}$$

$V(x)$: Finite potential well

$$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$$



- There are three regions – (I) well, (II) right of well, and (III) left of well.
- There are two cases – (i) total energy is less than potential barrier and (ii) total energy is greater than potential barrier
- Let us solve for case (i).

$V(x)$: Finite potential well – Case (i): $E < V_0$

Region I

- Kinetic energy is positive so

$$\psi_I(x) = A \exp(jk_I x) + B \exp(-jk_I x)$$

- Apply the **symmetry condition** of even or odd wave function

$$\text{Case (1): } A = B \quad \text{or} \quad \text{Case (2): } A = -B$$

Region II

- Kinetic energy is negative so

$$\psi_{II}(x) = C \exp(-\kappa_{II} x) + D \exp(+\kappa_{II} x)$$

- Apply the **finiteness condition**

$$\lim_{x \rightarrow \infty} \psi_{II}(x) = \text{finite}$$

$$\Leftrightarrow D = 0.$$

- Due to **symmetry** of potential well $\kappa_{III} = \kappa_{II}$.

Region III

- Kinetic energy is negative so

$$\psi_{III}(x) = E \exp(-\kappa_{III} x) + F \exp(\kappa_{III} x)$$

- Apply the **finiteness condition**

$$\lim_{x \rightarrow -\infty} \psi_{III}(x) = \text{finite}$$

$$\Leftrightarrow E = 0.$$

$V(x)$: Finite potential well – case (1): $A = B$

- Interface of Region I and Region II i.e. $x = \frac{+a}{2}$
-

Continuity condition of ψ

$$\begin{aligned}\psi_I\left(+\frac{a}{2}\right) &= \psi_{II}\left(+\frac{a}{2}\right) \\ \Leftrightarrow A \left[\exp\left(j\frac{k_I a}{2}\right) + \exp\left(-j\frac{k_I a}{2}\right) \right] \\ &= C \exp\left(-\frac{\kappa_{II} a}{2}\right) \\ \Leftrightarrow 2A \cos \frac{k_I a}{2} &= C \exp -\frac{\kappa_{II} a}{2} \quad (1)\end{aligned}$$

Continuity condition of $\frac{d\psi}{dx}$

$$\begin{aligned}\frac{d\psi_I}{dx}\left(+\frac{a}{2}\right) &= \frac{d\psi_{II}}{dx}\left(+\frac{a}{2}\right) \\ \Leftrightarrow jAk_I \left[\exp\left(j\frac{k_I a}{2}\right) - \exp\left(-j\frac{k_I a}{2}\right) \right] \\ &= -C\kappa_{II} \exp -\frac{\kappa_{II} a}{2} \\ \Leftrightarrow 2Ak_I \sin \frac{k_I a}{2} &= C\kappa_{II} \exp -\frac{\kappa_{II} a}{2} \quad (2)\end{aligned}$$

- Dividing (1) by (2) gives the **quantization condition** for wave vector.

$$\tan \frac{k_I a}{2} = \frac{\kappa_{II}}{k_I}$$

$V(x)$: Finite potential well – case (1): $A = B$

$$\tan \frac{k_1 a}{2} = \frac{\kappa_{II}}{k_1} \quad (3)$$

- This is a transcendental type of equation ³ and **cannot** be solved analytically!
- We have to employ numerical techniques.
- First let us introduce dimensionless variable z and constant z_0

$$z = \frac{k_1 a}{2} = \frac{a}{2} \sqrt{\frac{2mE}{\hbar^2}}, \quad z_0 = \frac{a}{2} \sqrt{\frac{2mV_0}{\hbar^2}}$$

so that (3) is reduced to

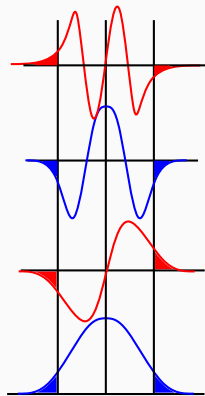
$$z \tan z = \sqrt{z_0^2 - z^2}.$$

- Now let us plot the graph of LHS and RHS of above equation. The points of intersection given you quantized values of z . From this, we can get quantized wave vectors and quantized energies.

³Transcendental means beyond conventional knowledge.

$V(x)$: Finite potential well – Nature of solutions

- Wave function penetrates into the finite potential barrier.
- The wave function is either odd or even function.
- The depth of the well determines the number of bound states.
- In the limiting case of very large depth, the penetration tends to zero and the number of bound states tends to infinity leading to particle in infinite potential well.



Key Insight

The wave function penetrates into the finite potential barrier. This is called **Quantum Penetration**. 